



### 3. Install Apache

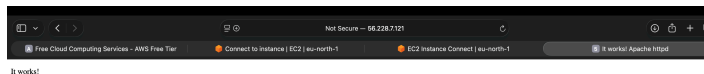
```
eu-north-1.console.aws.amazon.com

devops-cloud-challeng... Free Cloud Computing... eu-north-1 Instance Connect | eu-... Welcome to nginx!

AWS Search [Option+S] Ask Amazon Q

buntu@ip-172-31-20-191:~$ sudo apache2ctl configtest
syntax OK
buntu@ip-172-31-20-191:~$ sudo systemctl restart apache2
buntu@ip-172-31-20-191:~$ sudo systemctl status apache2
apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
Active: active (running) since Mon 2026-08-24 14:29:13 UTC; 1s ago
Invocation: 3caf1363c76c416584dfd605d1d162dc
Docs: https://httpd.apache.org/docs/2.4/
Main PID: 6545 (apache2)
Status: "Total requests: 0, Idle/Busy workers 100/0;Requests per second: 0.00 [0.000]
Tasks: 55 (limit: 627)
Memory: 5.6M (peak: 5.9M)
CPU: 57ms
CGroup: /system.slice/apache2.service
├─6545 /usr/sbin/apache2 -k start -DFOREGROUND
├─6548 /usr/sbin/apache2 -k start -DFOREGROUND
└─6549 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 24 14:29:13 ip-172-31-20-191 systemd[1]: Starting apache2.service: The Apache HTTP Server.
Aug 24 14:29:13 ip-172-31-20-191 systemd[1]: Started apache2.service: The Apache HTTP Server.
lines 1-17/17 (END)
```



### 4. Install nginx

```
eu-north-1.console.aws.amazon.com

*devops-cloud-challenges-22 (Cha... Free Cloud Computing Services - AW... Connect to Instance | EC2 | eu-north-1 EC2 Instance Connect | eu-north-1 Welcome to nginx!

AWS Search [Option+S] Europe (Stockholm) Rohithgoud (23116110641) Rohithgoud

Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.10) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

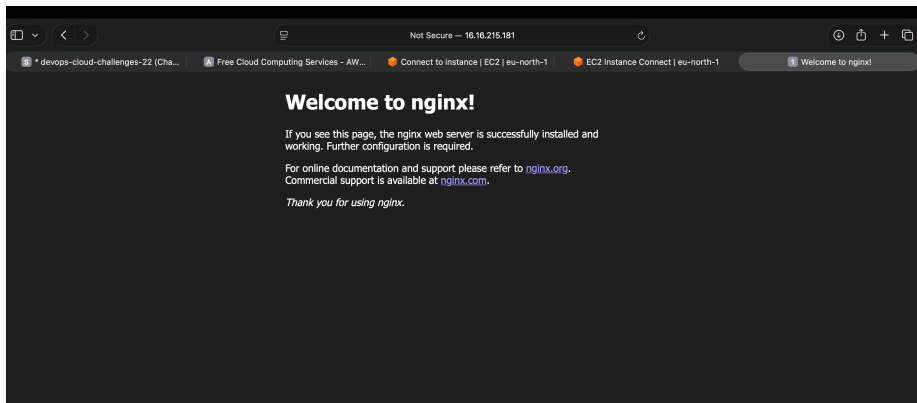
No services need to be restarted.

No containers need to be restarted.

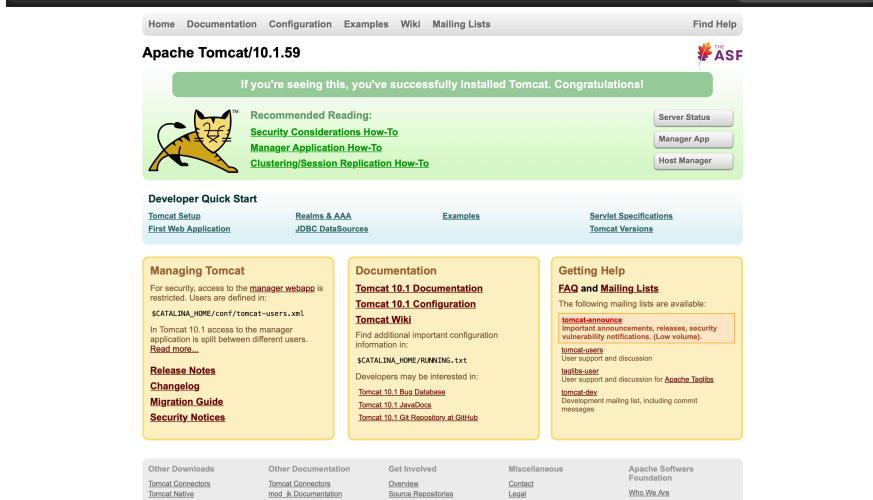
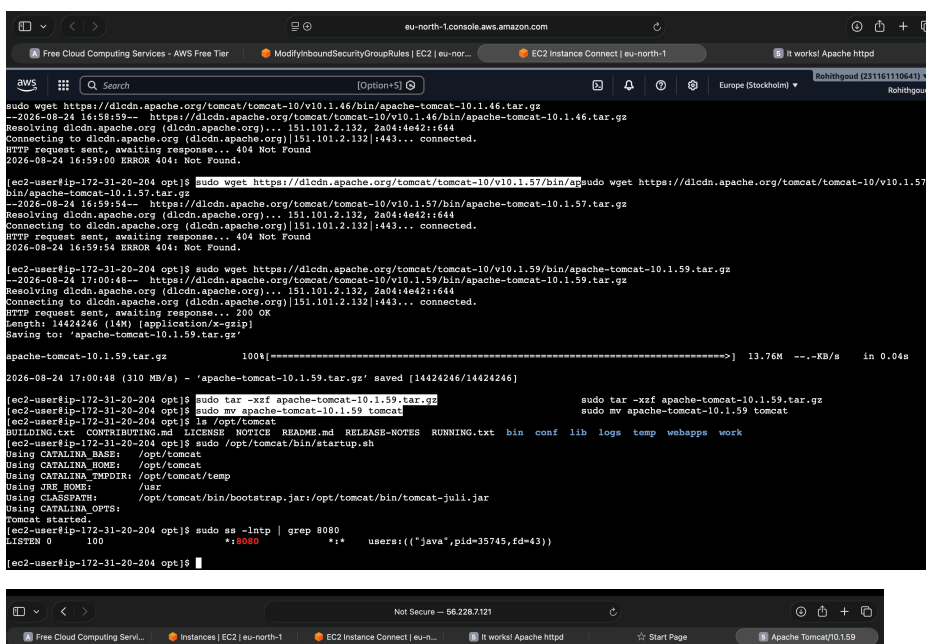
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
buntu@ip-172-31-20-191:~$ sudo systemctl start nginx
buntu@ip-172-31-20-191:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
buntu@ip-172-31-20-191:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Mon 2026-08-24 13:16:10 UTC; 1min 45s ago
Invocation: fd03975b958e434fa06d2c9cb6de9eb3
Docs: man:nginx(8)
Main PID: 2795 (nginx)
Tasks: 3 (limit: 627)
Memory: 3M (peak: 6.9M)
CPU: 41ms
CGroup: /system.slice/nginx.service
├─2795 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
├─2798 "nginx: worker process"
└─2799 "nginx: worker process"

Aug 24 13:16:10 ip-172-31-20-191 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 24 13:16:10 ip-172-31-20-191 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
buntu@ip-172-31-20-191:~$
```



## 5. Install Apache Tomcat



## 6.Install Docker.

```
eu-north-1.console.aws.amazon.com
devops-cloud-challenges-22 (Cha... AWS Management Console Connect to instance | EC2 | eu-north-1 EC2 Instance Connect | eu-north-1 (252) Winning Speech (Music Vid... Rehithgoud (231161110641) Rehithgoud

container-se-linux-4:2.245.0-1.amzn2023.noarch
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
pige-2.5-1.amzn2023.0.4.x86_64

containerd-2.2.5-1.amzn2023.0.2.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libnetfilterlink-1.0.1-19.amzn2023.0.2.x86_64
runc-1.3.5-1.amzn2023.0.2.x86_64

docker-25.0.16-1.amzn2023.0.4.x86_64
libbgroup-3.0-1.amzn2023.0.1.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-46-242 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-46-242 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-46-242 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-08-25 06:54:27 UTC; 50s ago
 TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
 Main PID: 20321 (dockerd)
    Tasks: 9
   Memory: 29.2M
      CPU: 387ms
   CGroup: /system.slice/docker.service
           └─20321 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal dockerd[20321]: time="2026-08-25T06:54:27.387397556Z" level=info msg="Starting up"
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal dockerd[20321]: time="2026-08-25T06:54:27.484457804Z" level=info msg="Loading containers: start."
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal dockerd[20321]: time="2026-08-25T06:54:27.913121367Z" level=info msg="Loading containers: done."
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal dockerd[20321]: time="2026-08-25T06:54:27.962702065Z" level=info msg="Docker daemon" commit="6fdf0a6"
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal dockerd[20321]: time="2026-08-25T06:54:27.962949680Z" level=info msg="Daemon has completed initialization"
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
Aug 25 06:54:27 ip-172-31-46-242.eu-north-1.compute.internal dockerd[20321]: time="2026-08-25T06:54:27.999588044Z" level=info msg="API listen on /run/docker.sock"
...skipping...
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-08-25 06:54:27 UTC; 50s ago
 TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
 Main PID: 20321 (dockerd)
    Tasks: 9
   Memory: 29.2M
      CPU: 387ms
   CGroup: /system.slice/docker.service
           └─20321 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536
```

## CHALLENGE .

Launch one EC2 instance using Amazon Linux 2.

Objective : Launch one EC2 instance using Amazon Steps Log in to the AWS Management Console. Navigate to EC2-> Instances -> Launch Instance. Enter the instance name: DevOps-Challenge. Select Amazon Linux 2 AMD AML. Select instance type: t3.micro. Create/select the required Key Pair. Create a new Security Group. Allow the required inbound ports: SSH-22 HTTP-80 Custom TCP-81 Custom TCP - 8080 Custom TCP -8081 Enable Auto-assign Public IP. Set the number of instances to 1. Click Launch Instance. Wait until the instance status becomes Running. Connect to the instance using Git Bash: ssh-i <key-file>.pem ec2-user@<Public-IP>

